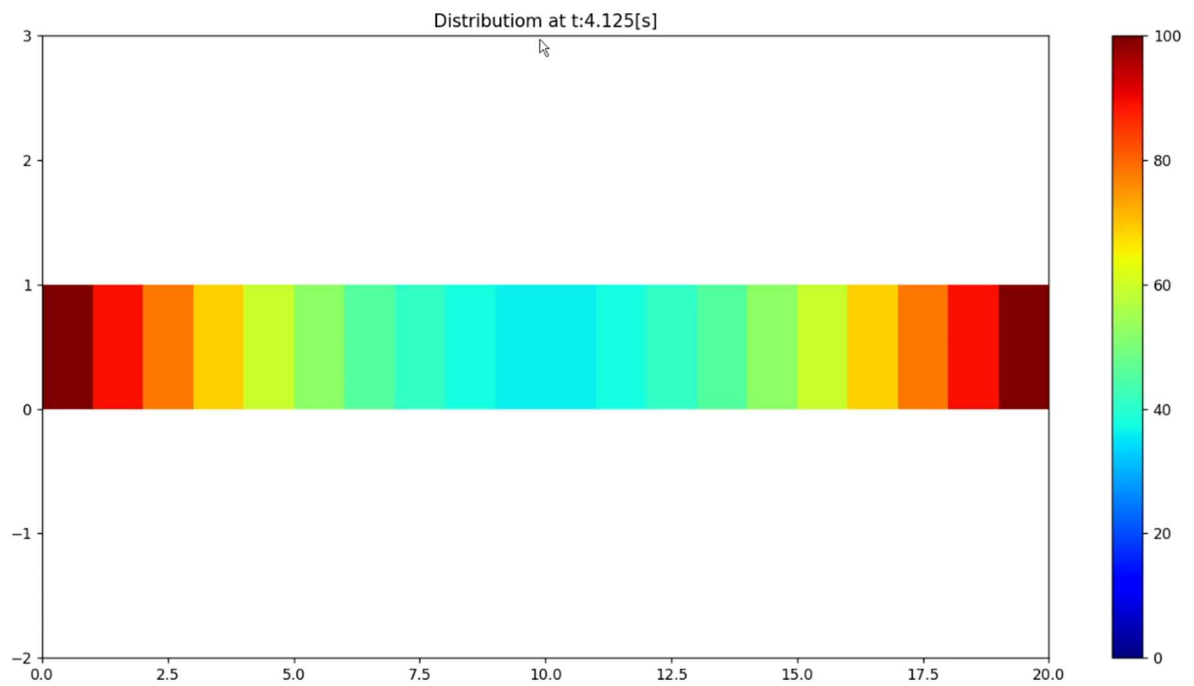
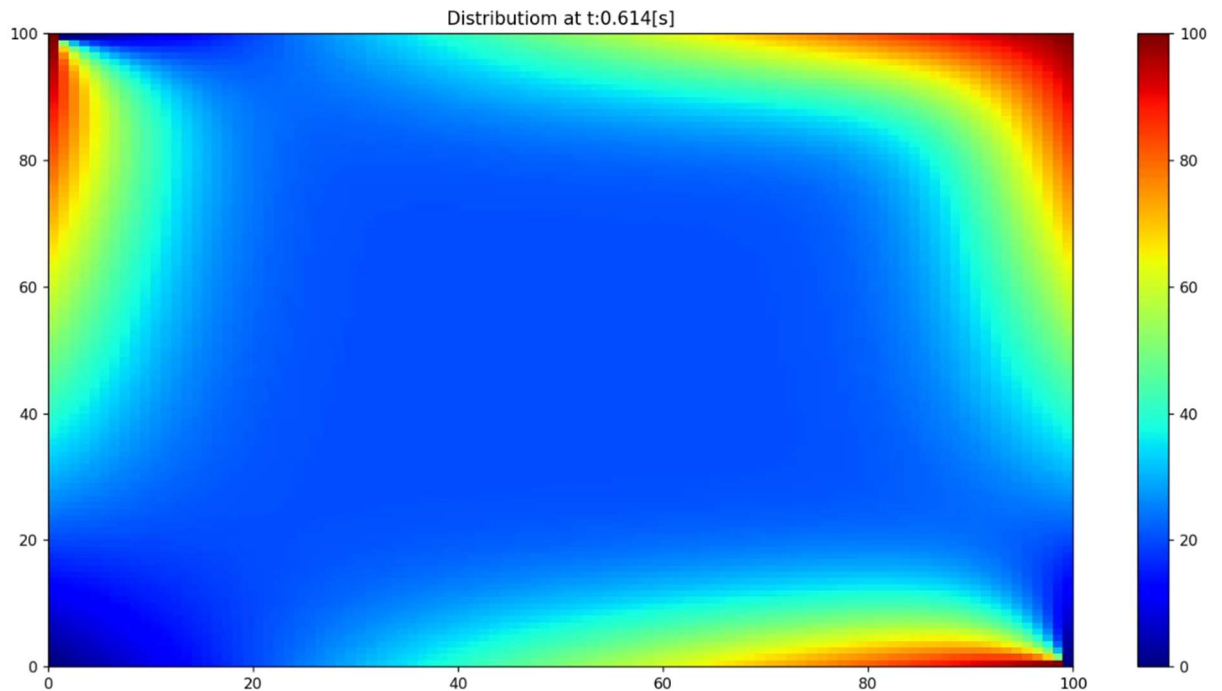


Python Projects:

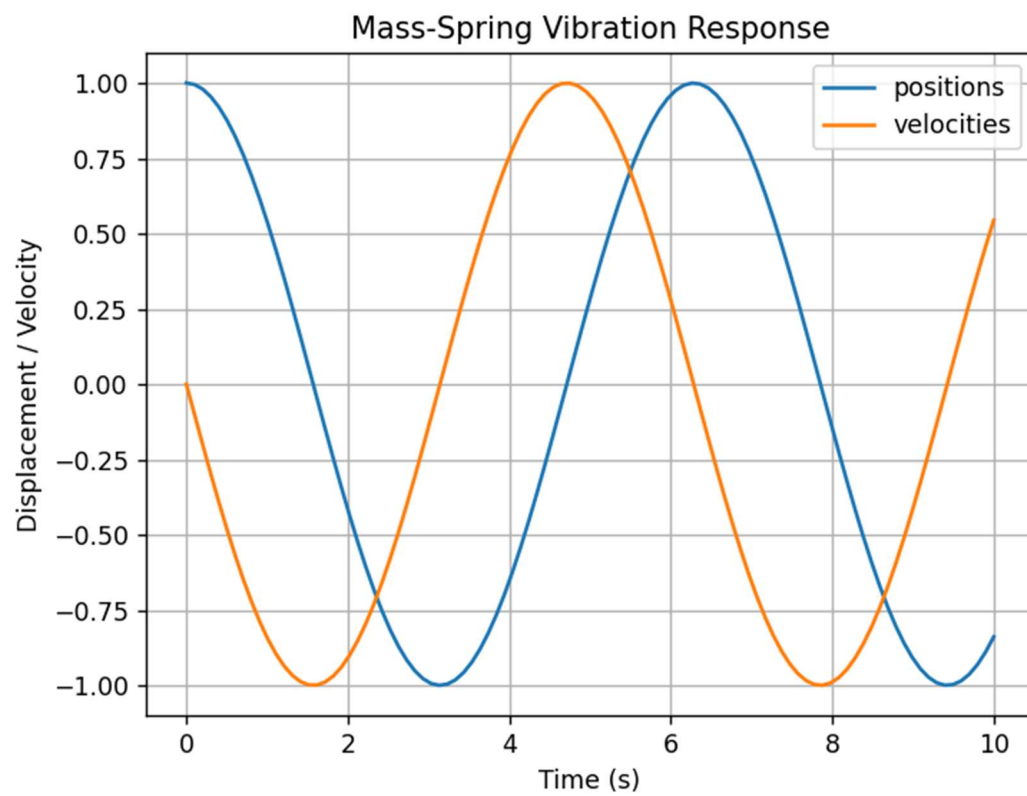
- ROD HEATED AT 2 ENDS (1D) SIMULATION :



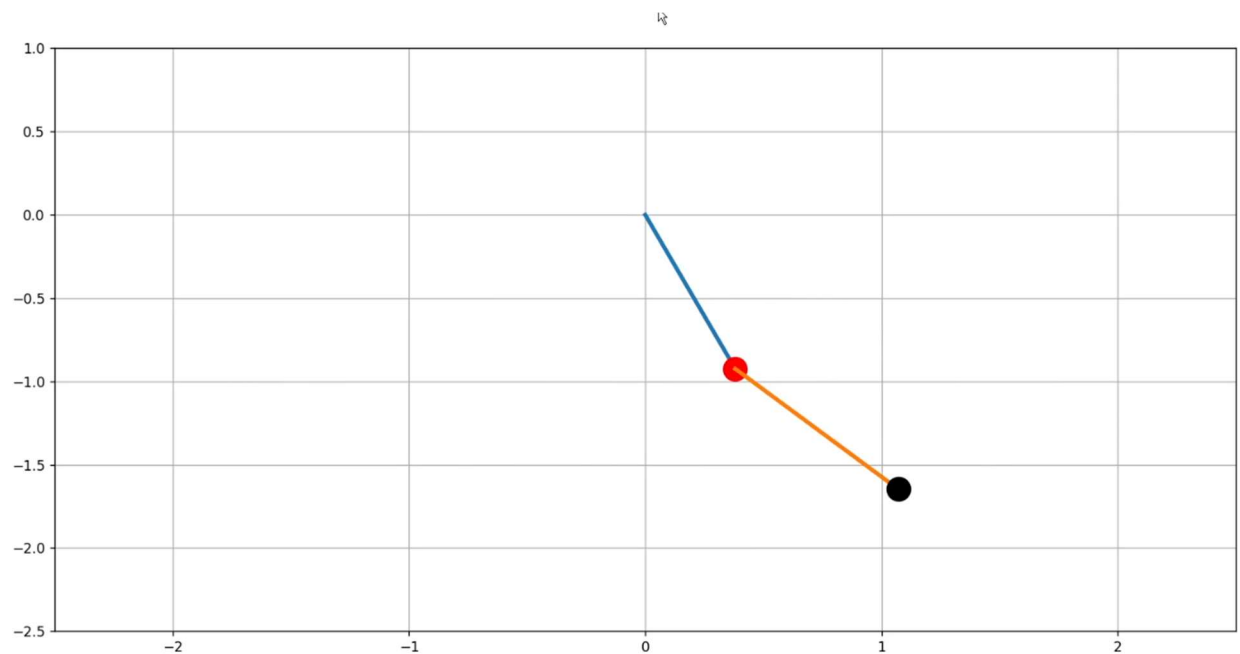
- PLATE HEATED AT DIFF PTS (2D) SIMULATION:



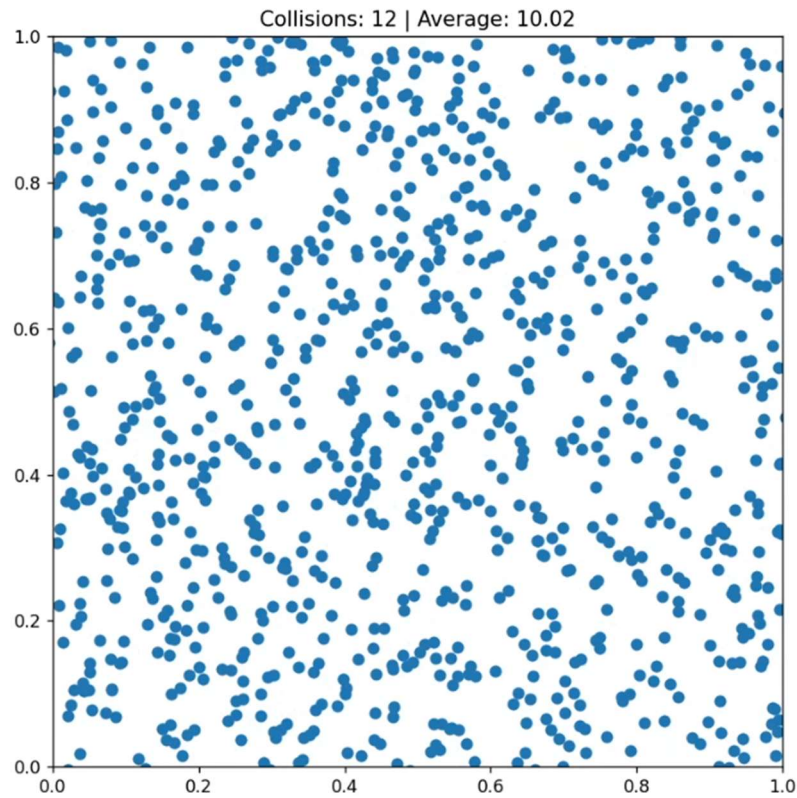
- MASS SPRING VIBRARTION MODEL



- DOUBLE PENDULUM SIMULATION



- UNIVERSAL GAS CONSTAT SIMULATION



- WEATHER DATA ANALYSIS

	Date/Time		Temp_C		Dew Point Temp_C		Rel Hum_%		Wind Speed_km/h		Visibility_km		Pressure	
	min	max	min	max	min	max	min	max	min	max	min	max	min	max
Weather														
Clear	1/11/2012 1:00	9/9/2012 5:00	-23.3	32.8	-28.5	20.4	20	99	0	33	11.3	48.3		
Cloudy	1/1/2012 17:00	9/9/2012 23:00	-21.4	30.5	-26.8	22.6	18	99	0	54	11.3	48.3		
Drizzle	1/23/2012 21:00	9/30/2012 3:00	1.1	18.8	-0.2	17.7	74	96	0	30	6.4	25.0		
Drizzle,Fog	1/23/2012 20:00	9/30/2012 2:00	0.0	19.9	-1.6	19.1	85	100	0	28	1.0	9.7		
Drizzle,Ice Pellets,Fog	12/17/2012 9:00	12/17/2012 9:00	0.4	0.4	-0.7	-0.7	92	92	20	20	4.0	4.0		
Drizzle,Snow	12/17/2012 15:00	12/19/2012 18:00	0.9	1.2	0.1	0.2	92	95	9	19	9.7	11.3		
Drizzle,Snow,Fog	12/18/2012 21:00	12/22/2012 3:00	0.3	1.1	-0.1	0.6	92	98	7	32	2.4	9.7		
Fog	1/1/2012 0:00	9/22/2012 0:00	-16.0	20.8	-17.2	19.6	80	100	0	22	0.2	9.7		
Freezing Drizzle	1/13/2012 10:00	2/1/2012 5:00	-9.0	-2.3	-12.2	-3.3	78	93	6	26	4.8	12.9		
Freezing Drizzle,Fog	1/1/2012 2:00	12/10/2012 5:00	-6.4	-0.3	-9.0	-2.3	82	94	6	33	3.6	8.0		
Freezing Drizzle,Haze	2/1/2012 11:00	2/1/2012 13:00	-5.8	-5.0	-8.3	-7.7	81	83	9	11	2.0	4.0		
Freezing Drizzle,Snow	1/13/2012 3:00	3/2/2012 12:00	-8.3	-3.3	-10.4	-4.6	79	94	6	24	2.4	12.9		
Freezing Fog	1/22/2012 6:00	3/17/2012 6:00	-19.0	-0.1	-22.9	-0.3	71	99	0	9	0.2	0.8		
Freezing Rain	1/13/2012 11:00	2/1/2012 7:00	-6.5	0.3	-9.0	-1.7	81	92	7	28	2.8	16.1		
Freezing Rain,Fog	1/17/2012 23:00	12/17/2012 1:00	-6.1	0.1	-8.7	-0.9	82	93	7	26	2.8	9.7		
Freezing Rain,Haze	2/1/2012 14:00	2/1/2012 15:00	-4.9	-4.9	-7.5	-7.4	82	83	6	9	2.0	2.8		
Freezing Rain,Ice Pellets,Fog	12/17/2012 3:00	12/17/2012 3:00	-2.6	-2.6	-3.7	-3.7	92	92	28	28	8.0	8.0		
Freezing Rain,Snow Grains	1/13/2012 9:00	1/13/2012 9:00	-5.0	-5.0	-7.3	-7.3	84	84	32	32	4.8	4.8		
Haze	1/22/2012 12:00	3/13/2012 23:00	-11.5	14.1	-16.0	11.1	68	86	0	17	4.8	9.7		
Weather Change	1/40/2012 11:00	9/6/2012 0:00	-29.9	33.0	-38.0	21.2	20	00	0	43	12.0	48.3		

THE STD DEVIATION IS: 0.844 AND THE VARIANCE IN REL HUMIDITY IS: 286.249

Number of exact 'Snow' records:

	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Pressure in Kilo Pascals	Weather
41	1/2/2012 17:00	-2.1	-9.5	57	22	25.0	99.66	Snow Showers
44	1/2/2012 20:00	-5.6	-13.4	54	24	25.0	100.07	Snow Showers
45	1/2/2012 21:00	-5.8	-12.8	58	26	25.0	100.15	Snow Showers
47	1/2/2012 23:00	-7.4	-14.1	59	17	19.3	100.27	Snow Showers
48	1/3/2012 0:00	-9.0	-16.0	57	28	25.0	100.35	Snow Showers
...	...	...	...	...	...	...	...	...
8779	12/31/2012 19:00	0.1	-2.7	81	30	9.7	100.13	Snow
8780	12/31/2012 20:00	0.2	-2.4	83	24	9.7	100.03	Snow
8781	12/31/2012 21:00	-0.5	-1.5	93	28	4.8	99.95	Snow
8782	12/31/2012 22:00	-0.2	-1.8	89	28	9.7	99.91	Snow
8783	12/31/2012 23:00	0.0	-2.1	86	30	11.3	99.89	Snow

[583 rows x 8 columns]

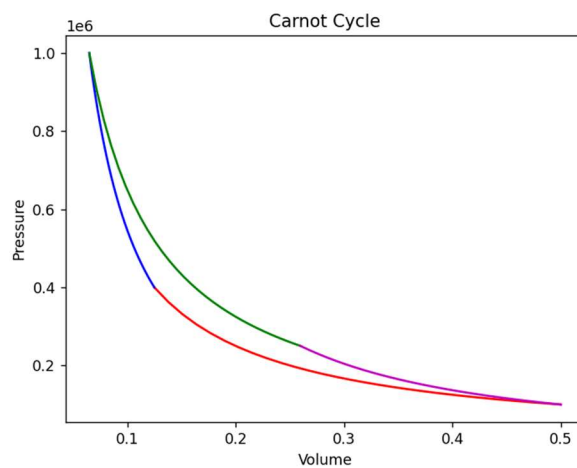
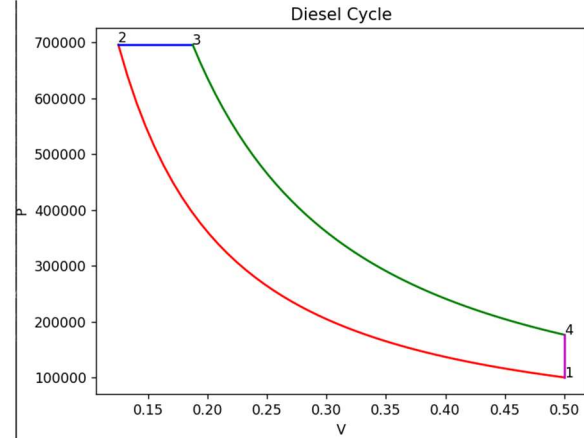
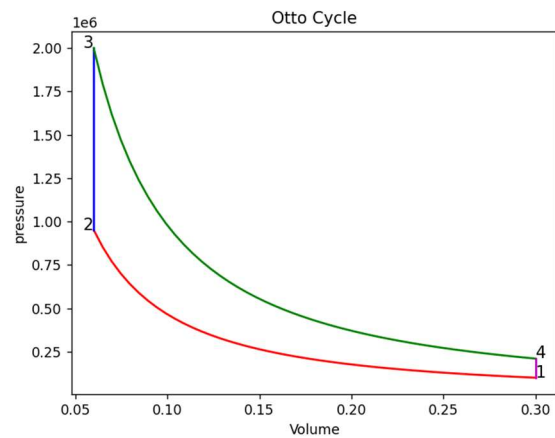
and the total number is 583

Number of times wind speed was above 24 and visibility was 25: 308

the rows are:

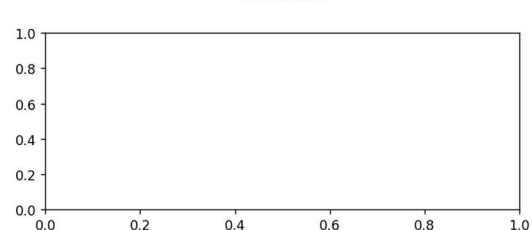
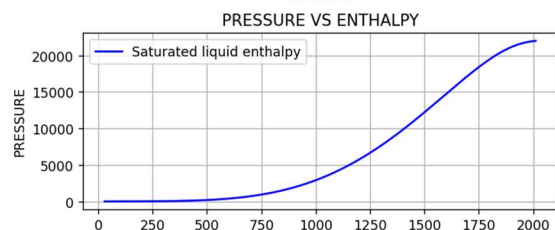
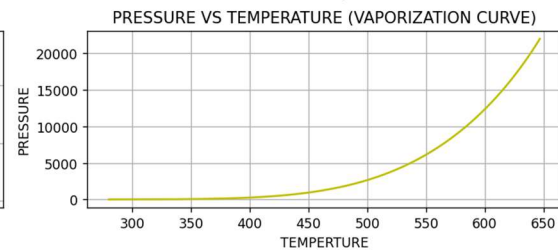
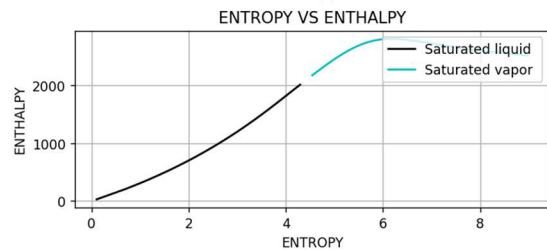
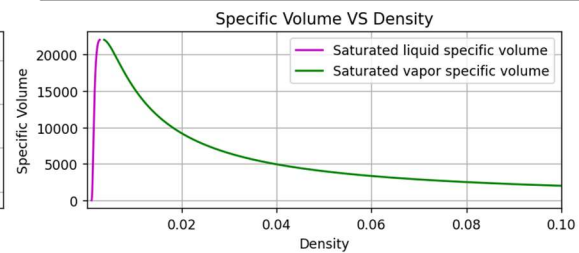
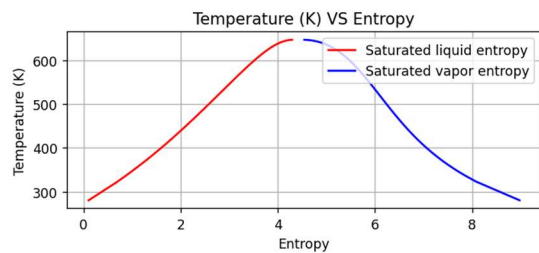
	Date/Time	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Pressure in Kilo Pascals	Weather
23	1/1/2012 23:00	5.3	2.0	79	30	25.0	99.31	Cloudy
24	1/2/2012 0:00	5.2	1.5	77	35	25.0	99.26	Rain Showers
25	1/2/2012 1:00	4.6	0.0	72	39	25.0	99.26	Cloudy
26	1/2/2012 2:00	3.9	-0.9	71	32	25.0	99.26	Mostly Cloudy
27	1/2/2012 3:00	3.7	-1.5	69	33	25.0	99.30	Mostly Cloudy
...	...	...	...	...	...	...	...	...
8705	12/28/2012 17:00	-8.6	-12.0	76	26	25.0	101.36	Partly Clear

- PLOTS FOR CARNOT, OTTO, DIESEL AND VAPOR DOME (+ STEAM VALUES)



```

THERMODYNAMIC PROPERTIES OF WATER :
Saturated liquid enthalpy : 1407.9564
Saturated vapor enthalpy : 2725.4874
Saturation temperature: 584.1474 K and in Celsius it is : 310.9974
Saturated liquid entropy : 3.3603
Saturated vapor entropy : 5.6159
Saturated liquid density: 688.6155
Saturated vapor density: 55.4682
Saturated liquid specific volume: 0.00145
Saturated vapor specific volume: 0.01803
    
```



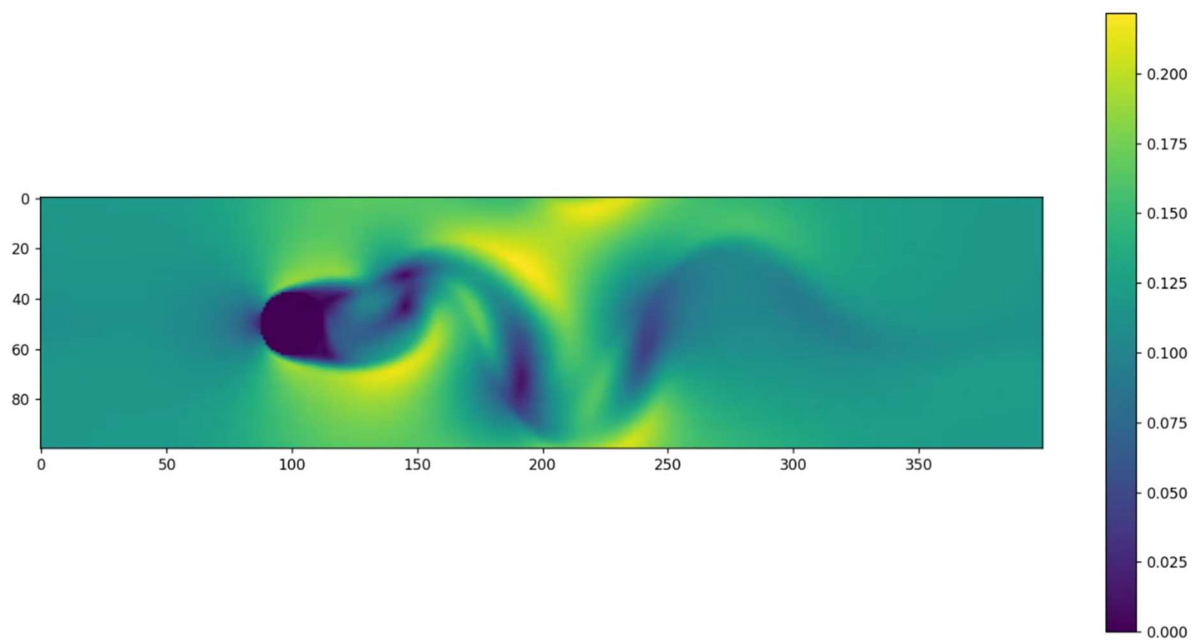
- LONDON HOUSING DATA ANALYSIS:

```
C:\Users\Reaga\PyCharmMiscProject\.venv\Scripts\python.exe "C:\Users\Reaga\PyCharmMiscProject\LONDON HOUSING analysis.py"
the number of data is :
date      13549
area      13549
average_price  13549
code      13549
houses_sold 13455
no_of_crimes 7439
dtype: int64
The number of null values in each column is:
date      0
area      0
average_price  0
code      0
houses_sold  94
no_of_crimes 6110
dtype: int64
```

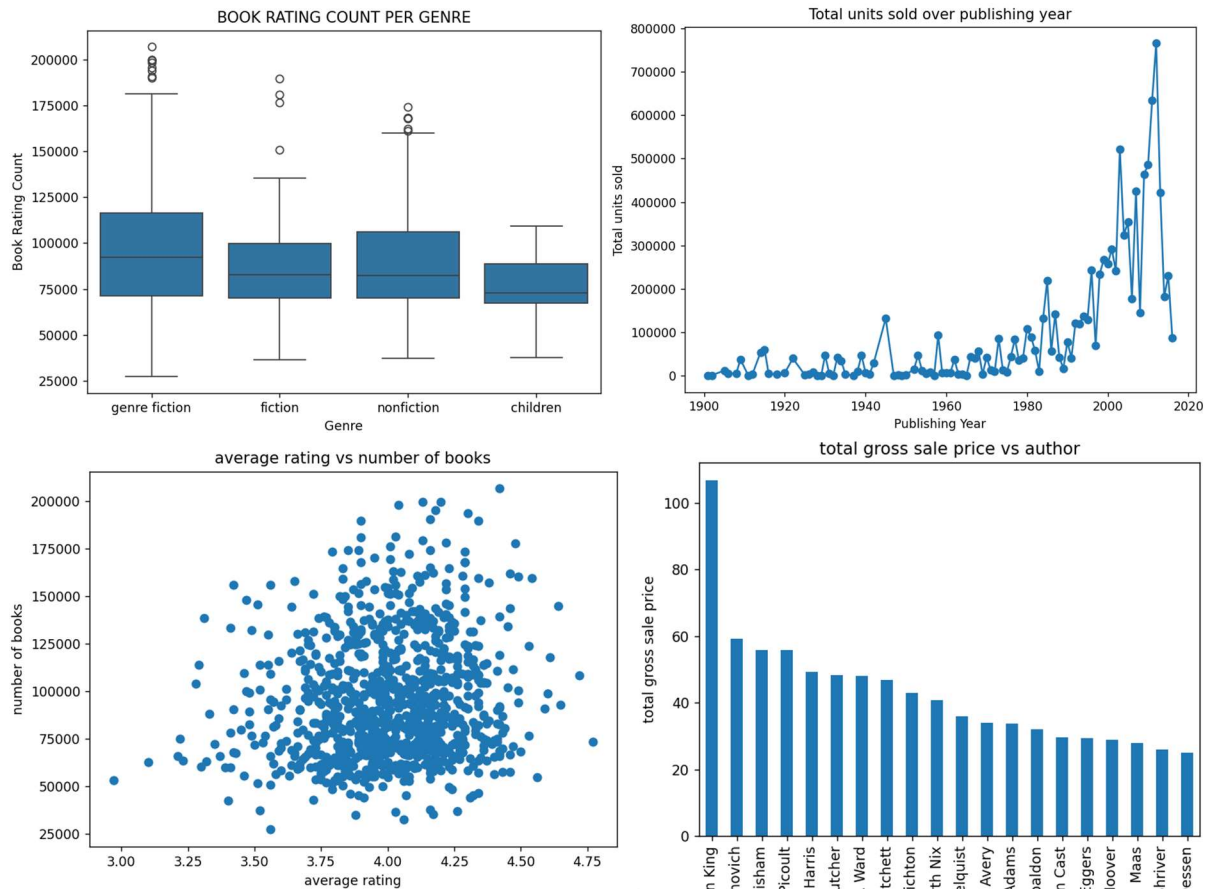
	date	area	average_price	code	houses_sold	no_of_crimes
0	1995-01-01	city of london	91449	E09000001	17.0	NaN
1	1995-02-01	city of london	82203	E09000001	7.0	NaN
2	1995-03-01	city of london	79121	E09000001	14.0	NaN
3	1995-04-01	city of london	77101	E09000001	7.0	NaN
4	1995-05-01	city of london	84409	E09000001	10.0	NaN

	date	area	average_price	code	houses_sold	no_of_crimes	year
0	1995-01-01	city of london	91449	E09000001	17.0	NaN	1995
1	1995-02-01	city of london	82203	E09000001	7.0	NaN	1995
2	1995-03-01	city of london	79121	E09000001	14.0	NaN	1995
3	1995-04-01	city of london	77101	E09000001	7.0	NaN	1995
4	1995-05-01	city of london	84409	E09000001	10.0	NaN	1995

- D2Q9 Lattice-Boltzmann Simulation



- BOOK SALES DATA ANALYSIS:



```
C:\Users\Reaga\PyCharmMiscProject\.venv\Scripts\python.exe "C:\Users\Reaga\PyCharmMiscProject\BOOK SALES analysis.py"
```

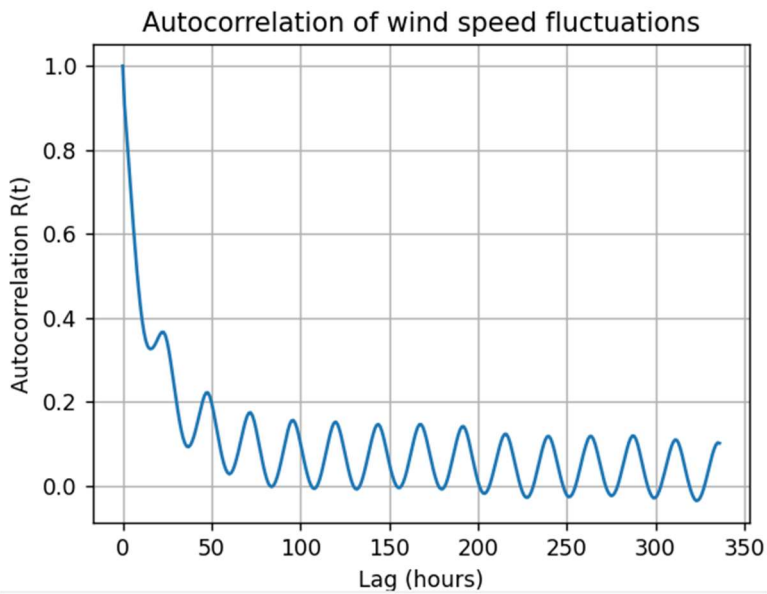
index	Publishing Year	Book Name	Author	language_code	Author_Rating	Book_average_rating	Book_ratings_count	gross sales	publisher revenue	sale price	sales rank	units sold
0	1975.0	Beowulf	Unknown, Seamus Heaney	en-US	Novice	3.42	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000
1	1987.0	Batman: Year One	Frank Miller, David Mazzucchelli, Richmond Lewis, Dennis O'Neil	eng	Intermediate	4.23	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000
2	2015.0	Go Set a Watchman	Harper Lee	eng	Novice	3.31	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000
3	2008.0	When You Are Engulfed in Flames	David Sedaris	en-US	Intermediate	4.04	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000
4	2011.0	Daughter of Smoke & Bone	Leini Taylor	eng	Intermediate	4.04	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000

index	Publishing Year	Book Name	Author	language_code	Author_Rating	Book_average_rating	Book_ratings_count	gross sales	publisher revenue	sale price	sales rank	units sold
0	1975.0	Beowulf	Unknown, Seamus Heaney	en-US	Novice	3.42	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000
1	1987.0	Batman: Year One	Frank Miller, David Mazzucchelli, Richmond Lewis, Dennis O'Neil	eng	Intermediate	4.23	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000
2	2015.0	Go Set a Watchman	Harper Lee	eng	Novice	3.31	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000
3	2008.0	When You Are Engulfed in Flames	David Sedaris	en-US	Intermediate	4.04	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000
4	2011.0	Daughter of Smoke & Bone	Laini Taylor	eng	Intermediate	4.04	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000

	index	Publishing Year	Book_average_rating	Book_ratings_count	gross sales	publisher revenue	sale price	sales rank	units sold
count	1070.000000	1069.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000	1070.000000
mean	534.500000	1971.377923	4.007000	94909.913084	1856.622944	843.281030	4.869561	611.652336	9676.980374
std	309.026698	185.080257	0.247244	31513.242518	3936.924240	2257.596743	3.559919	369.849830	15370.571306
min	0.000000	-560.000000	2.970000	27308.000000	104.940000	0.000000	0.990000	1.000000	106.000000
25%	267.250000	1985.000000	3.850000	70398.000000	372.465000	0.000000	1.990000	287.500000	551.250000
50%	534.500000	2003.000000	4.015000	89309.000000	809.745000	273.078000	3.990000	595.500000	3924.000000
75%	801.750000	2010.000000	4.170000	113906.500000	1487.957500	721.180500	6.990000	932.500000	5312.250000
max	1069.000000	2016.000000	4.770000	206792.000000	47795.000000	28677.000000	33.860000	1273.000000	61560.000000

	index	Publishing Year	Book_average_rating	Book_ratings_count	gross sales	publisher revenue	sale price	sales rank	units sold
count	1009.000000	1009.000000	1009.000000	1009.000000	1009.000000	1009.000000	1009.000000	1009.000000	1009.000000
mean	535.926660	1994.730426	4.012230	94817.793855	1832.644985	841.360638	4.844311	613.314172	9744.482656
std	308.769358	23.204719	0.246492	31473.890412	3947.885096	2279.579848	3.561712	369.628663	15350.021050
min	0.000000	1901.000000	2.970000	27308.000000	104.940000	0.000000	0.990000	1.000000	106.000000
25%	271.000000	1989.000000	3.860000	70701.000000	366.300000	0.000000	1.990000	291.000000	570.000000
50%	535.000000	2003.000000	4.030000	89204.000000	792.000000	273.240000	3.990000	596.000000	3942.000000

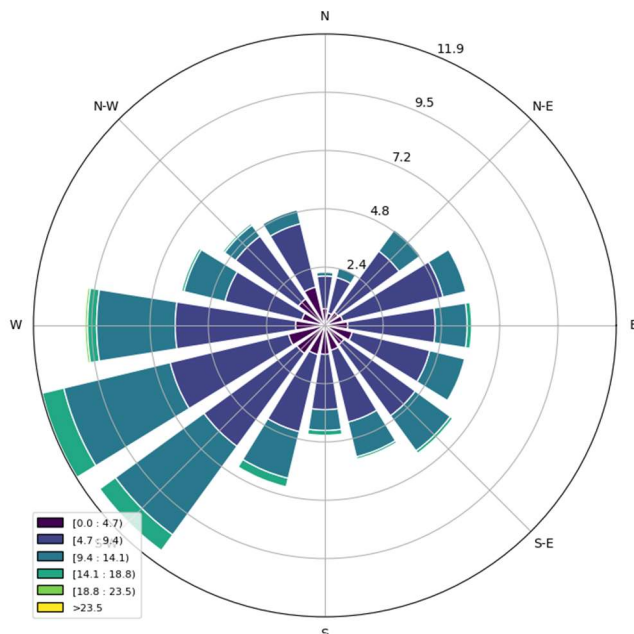
• WIND ANALYSIS:



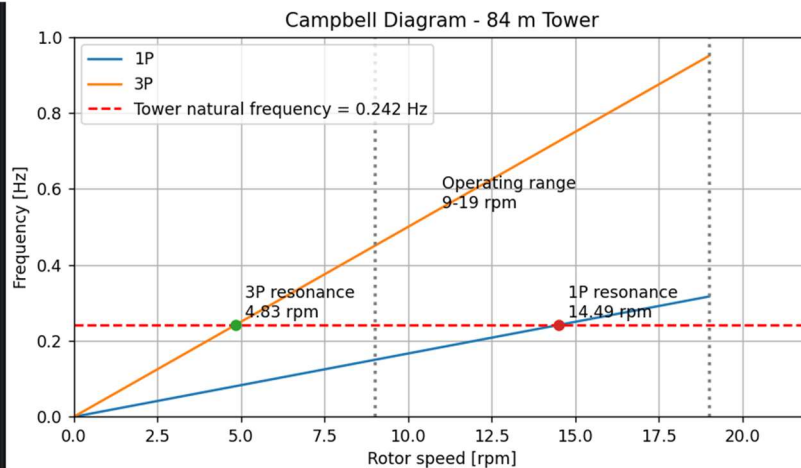
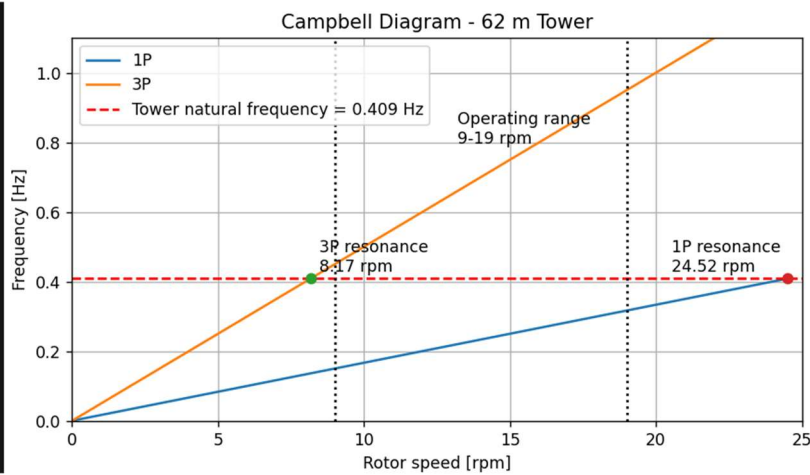
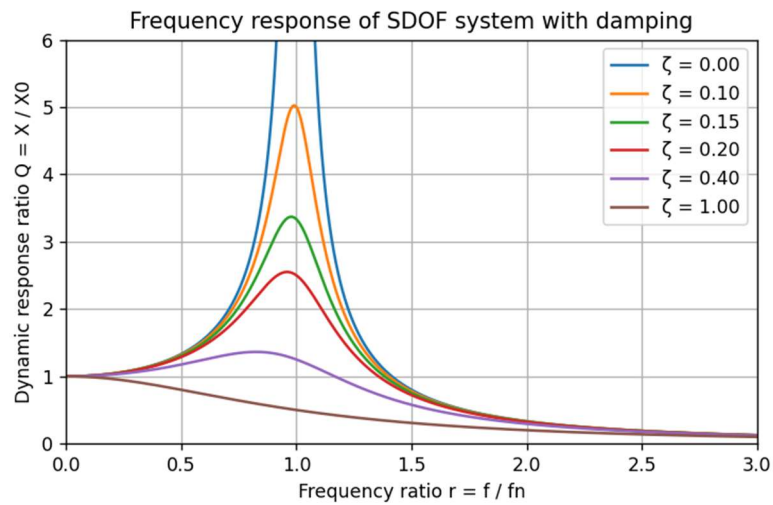
```
Name: wind_speed, dtype: float64
the mean is: 6.883436529523857e-17
The autocorrelation values are: [1.          0.90810939 0.85100823 0.79317592 0.73433148 0.67495432
 0.61749273 0.56278627 0.51202605 0.46645351]
The number of autocorrelation values are: 337
```

Lag (hours)	Wind speed (m/s)	Fluctuation $u(t)$ (m/s)	Autocorrelation $R(t)$
0	0	2.0	-1.238
1	1	1.3	-1.938
2	2	1.7	-1.538
3	3	2.0	-1.238
4	4	1.5	-1.738
5	5	1.3	-1.938
6	6	1.0	-2.238
7	7	1.5	-1.738
8	8	1.4	-1.838
9	9	2.0	-1.238
10	10	1.7	-1.538
11	11	1.7	-1.538
12	12	2.0	-1.238
13	13	3.0	-0.238
14	14	2.2	-1.038

The locations of the peaks on the lag axis are [23 48 72 96 120 144 168 192 216 240]
[25 24 24 24 24 24 24 24 24]
Average period = 24.111 hours
Integral time scale (days): 3.255

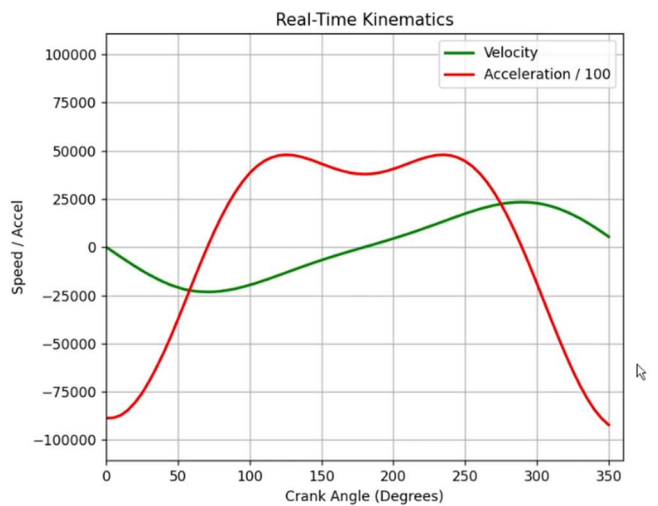
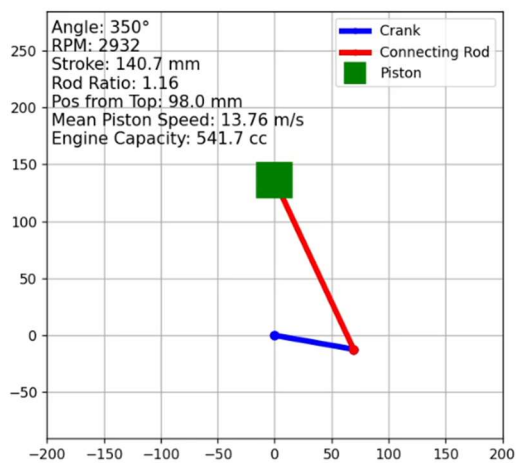
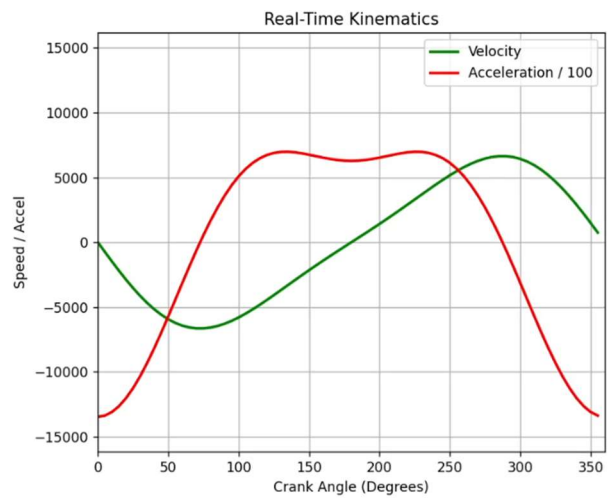
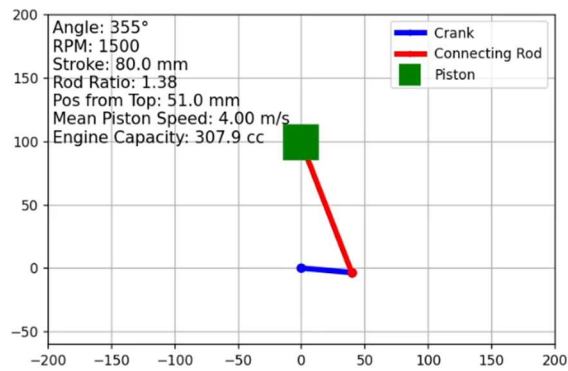


- VIBRATIONS IN A WIND TURBINE TOWER:



```
C:\Users\Reaga\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\Reaga\
Dynamic Response Ratio (Q) = 2.545567602784348
Displacement Amplitude (u0) = 0.021355704159807257 m
Natural frequency = 0.409 Hz
1P resonance speed = 24.52 rpm
3P resonance speed = 8.17 rpm
Natural frequency = 0.242 Hz
1P resonance speed = 14.49 rpm
3P resonance speed = 4.83 rpm
```

- Simulation of Engine Crankshaft



- Industrial Data Performance Improvement and Analysis

```
C:\Users\Reaga\PyCharmMiscProject\.venv\Scripts\python.exe "C:\Users\Reaga\PyCharmMiscProject\Industrial Energy & Production Performance Dashboard.py"

date shift cabinet_size_Kg order_qty good_units scrap_units downtime_breakdown_min downtime_changeover_min total_downtime_min machine1_breakdown
0 2026-01-01 1 8 kg 543 543 0 6 15 21 YES
1 2026-01-01 2 7 kg 413 395 18 21 12 33 YES
2 2026-01-02 1 7 kg 620 615 5 5 12 17 NO
3 2026-01-02 2 8 kg 556 536 20 25 15 40 YES
4 2026-01-05 1 8 kg 288 272 16 17 15 32 YES

=== OVERALL KPI SUMMARY ===
Total order quantity: 156916
Total good units: 154500
Total scrap units: 2416
Total downtime (min): 7541
Overall yield %: 98.46
Overall scrap %: 1.54

=== SHIFT-WISE KPIs ===
shift runs order_qty good_units scrap_units breakdown_min changeover_min total_downtime_min yield_pct scrap_pct avg_downtime_per_run
0 1 129 76674 75530 1144 1764 2004 3768 0.985080 0.014920 29.209302
1 2 129 80242 78970 1272 1811 1962 3773 0.984148 0.015852 29.248062

=== CABINET SIZE KPIs ===
cabinet_kg_num runs order_qty good_units scrap_units total_downtime_min yield_pct scrap_pct
0 7.0 98 58854 57973 881 2501 0.985031 0.014969
1 8.0 82 51008 50188 820 2411 0.983924 0.016076
2 10.0 78 47054 46339 715 2629 0.984805 0.015195

=== RISK LABEL SUMMARY ===
risk_label count
Normal 219
Warning 32
Critical 7

=== TOP FLAGGED RUNS ===
date shift cabinet_size_Kg order_qty good_units scrap_units total_downtime_min machine_breakdown_count anomaly_score risk_label
2026-06-26 2 10 kg 877 860 17 48 3 3 Critical
2026-04-14 1 10 kg 706 692 14 50 3 3 Critical
2026-03-31 1 10 kg 825 808 17 47 3 3 Critical
2026-05-07 1 10 kg 511 497 14 49 3 3 Critical
2026-04-22 2 10 kg 689 672 17 46 3 3 Critical
2026-02-04 1 10 kg 862 848 14 48 3 3 Critical
2026-02-13 2 10 kg 741 726 15 47 3 3 Critical
2026-03-13 1 10 kg 652 633 19 45 3 2 Warning
2026-01-28 1 8 kg 509 489 20 43 3 2 Warning
2026-04-27 1 10 kg 270 250 20 43 3 2 Warning
2026-01-19 2 8 kg 662 644 18 44 3 2 Warning
2026-05-08 2 8 kg 932 912 20 42 3 2 Warning
2026-06-05 1 10 kg 302 291 11 50 3 2 Warning
2026-05-22 2 10 kg 501 488 13 48 3 2 Warning
2026-05-04 2 10 kg 564 552 12 47 3 2 Warning
2026-02-17 2 10 kg 599 583 16 45 3 1 Warning
2026-06-24 1 8 kg 387 370 17 44 3 1 Warning
2026-03-02 1 10 kg 757 738 19 42 3 1 Warning
2026-01-21 2 8 kg 684 664 20 41 3 1 Warning

=== PART 3 OPTIMIZATION SUMMARY ===
Total scrap units before: 2416
Total scrap units after: 2174.4
Scrap units saved: 241.6
Total downtime before: 7541
Total downtime after: 6409.85
Downtime minutes saved: 1131.15
Good units before: 154500
Good units after: 154741.6
Extra good units gained: 241.6
Overall yield before %: 98.46
Overall yield after %: 98.61
Overall scrap before %: 1.54
Overall scrap after %: 1.39
Waste proxy reduction: 1372.75

=== SHIFT-WISE OPTIMIZATION SUMMARY ===
shift runs scrap_before scrap_after scrap_saved downtime_before downtime_after downtime_saved good_before good_after waste_proxy_before waste_proxy_after
1 129 1144 1029.6 114.4 3768 3202.80 565.20 75530 75644.4 4912 4232.40
2 129 1272 1144.8 127.2 3773 3207.05 565.95 78970 79097.2 5045 4351.85

=== CABINET SIZE OPTIMIZATION SUMMARY ===
cabinet_kg_num runs scrap_before scrap_after scrap_saved downtime_before downtime_after downtime_saved good_before good_after
7.0 98 881 792.9 88.1 2501 2125.85 375.15 57973 58061.1
8.0 82 820 738.0 82.0 2411 2049.35 361.65 50188 50270.0
10.0 78 715 643.5 71.5 2629 2234.65 394.35 46339 46410.5
```

